

Kyoto Protocol & Paris Agreement

Global Climate Action in the form of Treaties & Conferences

In this section, we will discuss some of the measures taken in the form of conferences and treaties that were formulated to mitigate the anthropogenic effects of global climate change. The earliest among these efforts was the treaty known as the United Nations Framework Convention on Climate Change (UNFCCC), followed by the Kyoto protocol, the annually conducted Conference of parties.

United Nations Framework Convention on Climate Change (UNFCCC)

The UNFCCC is an international treaty that was established to combat "dangerous human interference with the climate system". The UNFCCC was informally known as the Earth summit, and was held at Rio de Janeiro from 3-14 June 1992.

Its main objectives were to

- Stabilize greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system
- Set non-binding limits on greenhouse gas emissions for individual countries. It did not contain any enforcement mechanisms, due to which it was deemed unsatisfactory in reaching the required emission reduction goals.

Conference of the Parties (COP)

Conference of the Parties (COP) is the supreme decision-making body of the UNFCCC, which meets annually to assess progress in dealing with climate change. The first COP meeting was held in Berlin, Germany in March 1995. Usually, COP meets in Bonn, which is the seat of the secretariat, unless any member country offers to host the session. The latest COP was its 26th meeting, held from 31 October-12 November 2021. At COP-26, India pledged to become a net-zero carbon emitter by 2070 and announced enhanced targets for renewable energy deployment and reduction in carbon emissions.

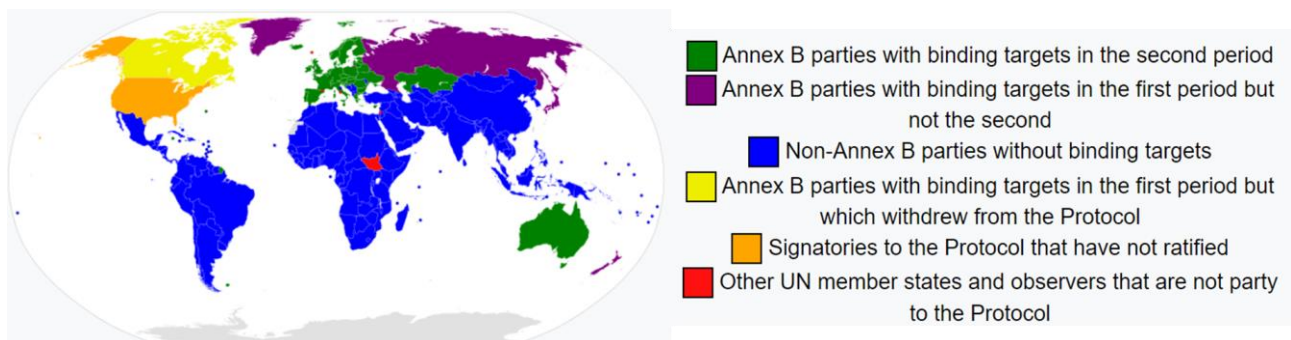
The Kyoto Protocol

Kyoto Protocol is an international treaty that was signed in 1997. It was the first implementation of measures formulated under the UNFCCC. It ran from the year 2005-2020. It was first adopted in Kyoto, Japan on the 11 December 1997 and entered into force on 16 Feb 2005; currently, there are 192 parties in the protocol. The Kyoto Protocol had two commitment periods, the first of which lasted from 2008 to 2012. It was amended again in 2012 to include the Doha amendment for the period 2013-2020.

Salient points of the Kyoto Protocol

- The Kyoto Protocol established three categories of signatory states, namely developed countries, developed countries with special financial responsibilities, and the third, developing countries.
- The developed countries, also called Annex 1 countries, originally consisted of 38 states, 13 of which were Eastern European states in transition to democracy and market economies, and the European Union.

- Annex II countries consisted of Developed countries with special financial responsibilities. E.g., Russia, Baltic states, Central and Eastern European states.
- Those countries not categorized under the Annex I and II countries were categorized as developing countries, including India and China, for example.



- The protocol was based on the principle of common but differentiated responsibilities: it acknowledged that individual countries have different capabilities in combating climate change, based on their economic development, and therefore placed the obligation to reduce current emissions on to developed countries on the basis that they are historically responsible for the current levels of greenhouse gases in the atmosphere. This included the Annex-I countries that were legally bound to lower their GHG emissions to 1990 levels. They were called upon to adopt national policies and take appropriate measures to mitigate climate change.

Goals of the Kyoto Protocol

The Kyoto Protocol applied to the seven greenhouse gases listed in carbon dioxide (CO₂), Methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF₆), nitrogen trifluoride (NF₃), which was added during the Doha amendment.

- Emission reduction targets were assigned for different countries, expressed as levels of assigned amount units (AAUs).
- The Kyoto protocol was legally binding, and any Annex-I or II country failing to meet targets were penalized.
- The US did not ratify the Kyoto Protocol, while Canada denounced it in 2012. The Kyoto Protocol was ratified by all the other Annex I Parties. All countries that remained parties to the Kyoto Protocol met their first commitment period targets.
- developing countries were required to only report their emissions to the UNFCCC.
- Flexibility mechanisms were established to help countries achieve their emission targets.

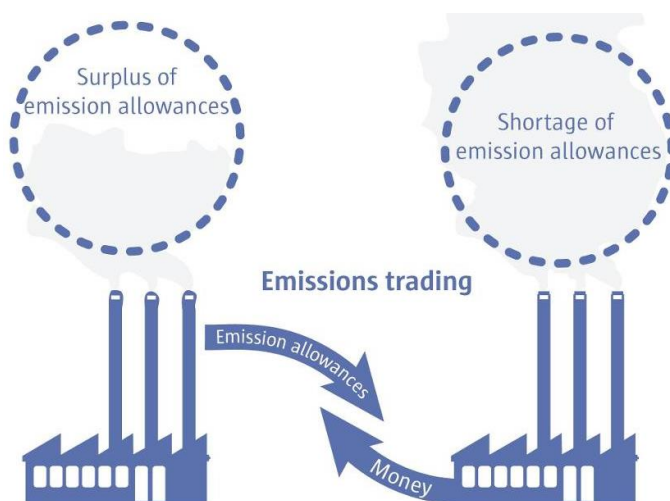
Flexibility Mechanisms of the Kyoto Protocol

The Protocol defines three "flexibility mechanisms" that can be used by the Annex I Parties in meeting their emission limitation commitments. These mechanisms aimed to lower the overall cost of achieving emission targets. The three mechanisms are:

- International Emissions Trading (IET)
- The Clean Development Mechanism (CDM)
- Joint Implementation (JI).

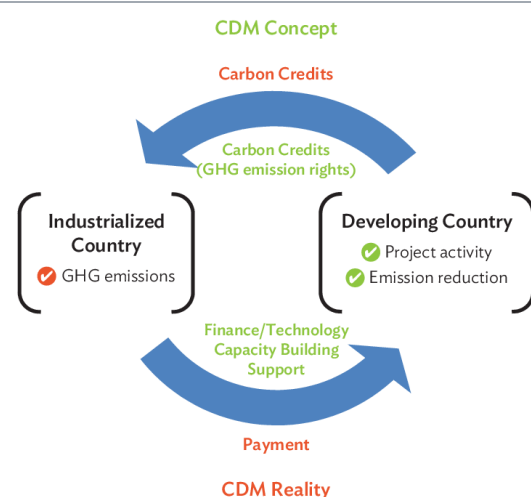
International Emissions Trading (IET)

International Emissions Trading (IET) allowed countries to trade unused emissions to other countries that exceeded their targets. For example, Country A has 100 emission units, of which it has used only 70. It can trade the remaining 30 units to another country B that has exceeded their permissible emission units. Thus, a new commodity was created in the form of emission reductions. Since CO₂ is the principle GHG, it is also referred to as carbon trading. Countries under the Kyoto protocol that were assigned targets for reducing their GHG emissions were expressed as levels of allowed emissions, or assigned amount units (AAUs). Emission trading currently operates across 35 countries in 4 continents.



Clean Development Mechanism (CDM)

Clean Development Mechanism (CDM) is a United Nations-run scheme that allows countries to fund GHG-reducing projects in other countries and claim the saved emissions as part of their own efforts to meet international emissions targets. These projects were also aimed at assisting developing countries achieve sustainable development and reduce their own carbon footprint. Here, let me introduce you to another form of carbon trading, known as Certified Emission Reductions (CER) units, also called as a carbon credit.



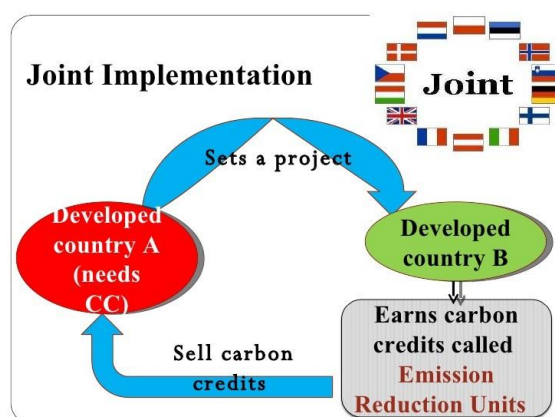
CDM = clean development mechanism, GHG = greenhouse gas.
Source: Asian Development Bank.

Carbon Trading

CER is a type of emission unit issued under the Clean Development mechanism to Annex-1 countries to help them comply with their emission reduction targets. CERs can be purchased either from a primary market (i.e., the country that makes the reduction) or from a secondary market (resold from a marketplace). CERs give the owner/country the right to emit 1 metric tonne of CO₂ or other equivalent GHG. CERs can be gained by developing projects that reduce GHG emissions.

Joint Implementation (JI) scheme

Under the Joint Implementation (JI) scheme, any Annex-I country could invest in a project to reduce GHG in any other Annex-I country as an alternative to reducing them domestically. This was introduced to lower the cost of reducing GHGs, as it may be expensive to do so in certain countries and cheaper in others. Under the JI scheme, another form of



carbon trading, known as the Emission Reduction Unit (ERU) was introduced. 1 ERU represents the reduction of 1 tonne of CO₂.

HOW CARBON TRADING WORKS



Carbon credits are received from auctions, or given for free to firms by governments



Carbon credits are used as greenhouse gases are emitted



Unused carbon credits can be sold, and more credits can be bought

Paris Agreement

The Paris agreement, or the Paris Climate Accords is an international treaty on climate change that was adopted in 2015. It covers climate change mitigation, adaptation and finance. The agreement was negotiated by 196 parties at the 2015 UNFCCC near Paris, France. It opened for signature on 22 April 2016 (Earth Day), and entered into force on 4 November 2016. As of November 2021, 193 members of the UNFCCC ratified the Paris agreement, and only 4 countries have not ratified, of which Iran is the major emitter. USA withdrew from the agreement in 2020, but re-joined in 2021.



The long-term goal of the Paris agreement was to restrict the rise in mean global temperature to <2°C, and if possible, to 1.5°C. to achieve this goal, it has been estimated that emissions need to be cut by 50% by 2030. No specific emission targets were enforced on countries like in the case of the Kyoto protocol, but it was mandated that targets should not exceed the previous ones set out by the Kyoto protocol.